

Modularity Maximization: An Experimental Survey

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Abstract

There has been a wide variety of community detection methods proposed over the last two decades that optimize the modularity objective. Despite this abundance of methods, comparative experimental evaluations remain fragmented: most studies consider only a small subset of available algorithms. As a result, it is often unclear how different modularity-based methods relate to one another in practice, or under which conditions specific design choices are most effective. Most notably, the body of work can largely be divided in two main categories based on whether they use machine learning or not.

In this work, we present an experimental survey of modularity optimization methods, covering more than twenty representative approaches spanning different algorithmic paradigms. We summarize key algorithmic features, and provide a systematic experimental comparison across multiple datasets and evaluation criteria. Our goal is not to identify a single “best” method, but to clarify empirical trade-offs, expose gaps in existing comparisons, and offer practical guidance for researchers and practitioners working with modularity-based graph clustering.

CCS Concepts

• **Theory of computation** → **Graph algorithms analysis**; • **Mathematics of computing** → **Graph theory**; • **Computing methodologies** → **Cluster analysis**.

Keywords

Community detection, Graph clustering, Modularity optimization, Experimental evaluation

ACM Reference Format:

Adil Chhabra, Kenneth Langedal, and Christian Schulz. 2018. Modularity Maximization: An Experimental Survey. In *Proceedings of Make sure to enter the correct conference title from your rights confirmation email (Conference acronym 'XX)*. ACM, New York, NY, USA, 3 pages. <https://doi.org/XXXXXX.XXXXXXX>

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Conference acronym 'XX, Woodstock, NY

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ACM ISBN 978-1-4503-XXXX-X/2018/06
<https://doi.org/XXXXXX.XXXXXXX>

1 Introduction

Community detection in graphs.

- Problem definition
- Why modularity?
- State why this survey is needed, missing comparisons in the literature
- At the end, list of contributions (experimental survey stuff)

2 Preliminary

We use the term algorithm to denote a concrete computational procedure. Except where explicitly stated, the algorithms considered here are heuristics for modularity maximization and do not guarantee optimality.

3 Related Work

4 Methods

This section contains a short description outlining the core ideas behind each clustering method. They are grouped the algorithms into categories to make it easier to navigate the survey. With in each category, the algorithms are presented in order of publication date. At the top level of this taxonomy, we have *Learning-based* methods and *Classical modularity optimization* methods. For this distinction, *Learning-based* refers to any method that train model parameters as part of the algorithm.

4.1 Classical Modularity Optimization Methods

CNM. A scalable greedy modularity-optimization algorithm with near-linear running time on large sparse networks and demonstrates its effectiveness on a large real-world network.

Walktrap. Based on the idea that random walks on a graph tend to get “trapped” into densely connected parts corresponding to communities.

Infomap. Infomap frames community detection as an information compression problem: it partitions a graph so that the expected code length of a random walk on the network is minimized, thus uncovering modules where flows concentrate. The alternative *Map Equation* objective function was also introduced here.

WT. TODO.

Louvain. The famous Louvain method, named after the location the work took place, is a multi-level approach to graph clustering. At each level, the algorithm greedily moves vertices to neighboring clusters based on the modularity delta. Once there are no improving moves, each cluster is contracted into a single vertex, and the process repeats.

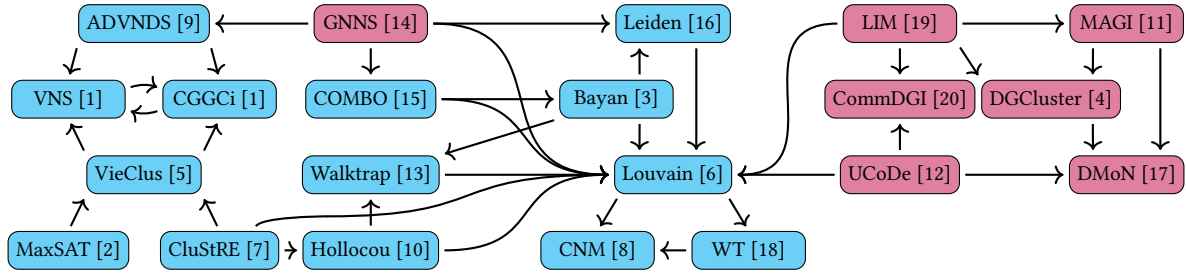


Figure 1: Previous experimental comparisons visualized. Each directed edge indicates that the source algorithm used the destination algorithm in the experimental evaluation. The colors indicate either a learning-based method when red, or a classical optimization algorithm when blue.

Table 1: Comparison of the methods included in the experimental survey. The *Code Available* field is defined as follows: $\times$ means not available, one $\star$ means code is available but not prepared for use in any way, $\star\star$ is for methods with open source implementations with well defined input/output formats, and $\star\star\star$ means the algorithm is easy to use via portable Python packages such as the CDlib package.

Method	Year	Code Available	Stochastic	Learning-Based	Fixed Cluster Count	Exact	Approx. Running Time
CNM	2004	$\star\star\star$					$O(n \log^2 n)$
Walktrap	2005	$\star\star\star$	$\checkmark$				$O(n^2 \log n)$
WT	2007	$\star\star\star$					Improved CNM
Louvain	2008	$\star\star\star$	$\checkmark$				?
CGGCI	2012	$\times$	?				?
VNS	2012	$\times$	$\checkmark$				?
COMBO	2014	$\star\star\star$	?				?
ADVNDs	2017	$\times$	$\checkmark$				Search-based
HollocoU	2017	$\star\star$	?				?
VieClus	2018	$\star\star$	$\checkmark$				?
Leiden	2019	$\star\star\star$	$\checkmark$				?
DMoN	2020	$\star$	?	$\checkmark$		$\checkmark$	?
CommDGI	2020	$\star$	?	$\checkmark$		$\checkmark$	?
GNNs	2022	$\star$	?	$\checkmark$			?
UCoDe	2023	$\star$	?	$\checkmark$		$\checkmark$	?
DGCluster	2023	$\star$	?	$\checkmark$			?
MAGI	2024	$\star$	?	$\checkmark$		$\checkmark$	?
Bayan	2024	$\star\star$				$\checkmark$	?
CluStRE	2025	$\star\star$	?				?
LIM	2025	?	?	$\checkmark$		?	?
MaxSAT	2025	?				$\checkmark$	?

CGGCI-RG. Combination of the CGGC (Core Groups Graph Clustering) ensemble scheme with an underlying Randomized Greedy (RG) clustering algorithm. It was the best performing algorithm submitted to the 10th DIMACE Implementation Challenge.

VNS. Variable Neighborhood Search metaheuristic that obtained the second place in the 10th DIMACS Implementation Challenge. It was a close call between VNS and CGGCI-RG.

COMBO. TODO.

HollocoU. Modularity optimization in the streaming setting. For each streamed edge, HollocoU employs a simple decision strategy

based on a given parameter and the volumes of the communities of the two nodes.

VieClus. A general memetic algorithm that can be adapted to optimize different objective functions.

Leiden. As a direct improvement to the Louvain method, Leiden addresses a defect in the Louvain method that can lead to disconnected communities.

Bayan. An exact algorithm for modularity optimization that uses Integer Linear Programming solvers.

CluStRE. Improved streaming algorithm with both re-streaming and combined use of VieCLUS in-memory after streaming and contraction.

MaxSAT. An improved exact method for modularity optimization using a MaxSAT formulation.

4.2 Learning-Based Methods

Methods that find model parameters.

DMoN.

CommDGL.

UCoDe.

DGCluster.

MAGI.

5 Experiments

6 Conclusion

Acknowledgments

To Robert, for the bagels and explaining CMYK and color spaces.

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A Research Methods

A.1 Part One

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Received 20 February 2007; revised 12 March 2009; accepted 5 June 2009